

Feb 19, 2026

Introduction to Computer and Network Security

Part 1

Question 1 [20 points]

Describe the CIA triad properties along with the authenticity and non-repudiation properties; give for each one of them, an example.

Question 2 [30 points]

In the context of cryptography, (a) state and discuss the Kerckhoffs' principle, (b) explain the purpose of key management, (c) explain the differences between symmetric and asymmetric cryptography.

Part 2

You have been hired by a news agency which is starting to provide an AI-powered chatbot able to answer readers' questions, with the purpose of helping them understand some technical details and the overall impact on their security and privacy posture. You are supposed to help them assess the quality of the answers.

Question 3 [50 points]

You read the following statement:

Cadabra is a company that sells home automation IoT devices. The offered devices upload all the pictures, voice recordings and logs into a database stored within the company's free cloud storage. Each user has their own isolated database which is only accessible by the user and Cadabra itself, for this reason, the stored information is transmitted via HTTP and stored without any encryption.

To assess the chatbot reliability on cross-checking information, you act as a concerned user and have the following interactions:

1.	You: What does it mean that the "information is stored without any encryption"?
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	Chatbot: It means that the information is hashed locally by using the device's public key, and then uploaded as digests. This ensures that the information can only be read by inverting the function.
2.	You: What does it mean that the "information is transmitted via HTTP"? Chatbot: It means that the communication channel shared by your devices and their server ensures both confidentiality and integrity. For this reason, the original statement is wrong as your data are safe.
3.	You: Is the Cadabra infrastructure GDPR-compliant in the US? Chatbot: I am sorry, the statement does not contain enough information for me to answer your question.
4.	You: Can Cadabra employees access my personal information if they want to? Chatbot: No, they cannot as Cadabra follows the Principle of Least Privilege, and thus they can only read your plaintext information if they don't have any privilege on them.

Describe the possible mistakes the chatbot has made in each one of the four interactions (use the numbers for the explanations), detailing how the mentioned mechanisms work, their features and actual impact.

Please, remember...

- on each used sheet of paper (possibly, 2 sheets at most), annotate:
 - first and last name
 - student ID number
 - question number
- write either in Italian or English, acronyms do not need to be translated;
- only pens are allowed. Drafts and text written using pencils will not be considered during the exam revision;
- adequately justify your answer. A phrase and a paragraph-long answer will have a significantly different grade;
- write clearly, your answers will not be *interpreted*;
- after turning in your exam, please respect your classmates by leaving quietly.